

Design of Rectangular Microstrip 4x2 Patch Array Antenna at 2.4 GHz for WLAN Application

Amitha.R.Nair

Bharati A. Singh

S.S. Thakur

Electronics & Telecommunication
Shree L.R. Tiwari College of Engg.
Thane, India
nair.amitha@gmail.com

Electronics & Telecommunication
K.J.Somaiya College of Engg.
Mumbai, India
bhartisingh@somaiya.edu

Electronics & Telecommunication
Shree L.R. Tiwari College of Engg.
Thane, India
sanjayami@gmail.com

Abstract— An array of Rectangular Microstrip Antenna (RMSA) is designed and compared with single patch RMSA and optimized for observing an increase in directivity along with achieving expected radiating parameters such as $S_{11} < -10\text{dB}$ and Side Lobe Level (SLL) $< -20\text{dB}$ using IE3D simulator. Maximum directivity of 13.058dBi is obtained by implementing four by two (4x2) array of eight patches as compared to 6.74dBi for single patch for the application of WLAN at 2.4 GHz centre frequency. The parametric variations are done for obtaining desired results.

Keywords— RMSA, Four by Two Array (4x2), IE3D, FR4.

I. INTRODUCTION

An antenna plays a very important role in wireless communication systems. For applications emerging of compact size technologies, microstrip antennas are used because of their low cost, portability, versatile, conformal and other factors supporting the needs. But microstrip antennas also have disadvantages like narrow bandwidth, low gain and excitation of unwanted surface waves. Hence antenna arrays are used to overcome for these disadvantages, thereby introducing microstrip patch arrays. This technology of microstrip arrays makes it possible to provide high gain, desired radiation pattern, appropriate beamwidth with least cost and less space. Recent advances have shown increase in the bandwidth and gain using microstrip array [1].

For desired radiation pattern and directivity each element in the array can be fed in many ways ; a) same amplitude and same phase difference b) same amplitude and varying phase difference c) varying amplitude and same phase difference and d) varying amplitude and varying phase difference. But uniform arrays are used which are easy for construction [2], [4].

In this paper four by two (4x2) array, edge fed by a microstrip line for WLAN at 2.4 GHz is presented using inset feeding method in contrast to reference [2] where a four by one (4x1) array is designed for WiMAX application at 2.5 GHz.

Quarter wave transformers are used to match the impedance of the transmission lines considering SMA connector at the input to the antenna [4]. The resonant frequency is determined to be 2.4 GHz.

Using IE3D simulator, the optimized antenna array of 4x2 patches gives a return loss (S_{11}) of -19.64dB. Many optimizations are carried out by for getting better expected results for gain and directivity. Next part explains the antenna designing procedure respectively.

II. ANTENNA DESIGN

Starting with the designing of a rectangular microstrip patch antenna, there are some parameters chosen. Those are substrate related parameters such as its relative permittivity, loss tangent and height of the substrate. Available dielectric lies in the range from $2.2 < \epsilon_r < 12$, and lower permittivity is chosen for better performance [2] in terms of gain. FR4 substrate is chosen having ϵ_r as 4.3 and the loss tangent as 0.01 with 1.6mm as thickness as it is easily available.

A single patch is designed for the given application and selected substrate. And consequently 4x2 array is designed for comparing its performance with that of single patch. Table I depicts the material specification selected for the design and application.

TABLE I DESIGN SPECIFICATION FOR RECTANGULAR PATCH ANTENNA

Centre Frequency	2.4 Gz
Substrate	FR4
Dielectric Constant ϵ_r	4.3
Substrate height	1.6mm
Loss Tangent ($\tan \delta$)	0.01

A. Single patch antenna design dimension:

The objective of this part is to design parameters for a single patch, feedline specifications and also the quarter wave transformer length. Transmission line method is used for designing [4]. The width W and length L is calculated as follows:

$$W = \frac{c}{2f_r} \sqrt{2/(\epsilon_r + 1)} \quad (1)$$

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \sqrt{\left(1 + 12 \frac{h}{W}\right)} \quad (2)$$

$$L_1 = \frac{c}{2 f_r \sqrt{\epsilon_r}} \quad (3)$$

$$\Delta L = 0.412 h \left(\frac{\epsilon_{eff} + 0.3}{\epsilon_{eff} - 0.258} \right) \left(\frac{(W/h) + 0.264}{(W/h) + 0.8} \right) \quad (4)$$

$$L = L_1 - 2 \Delta L \quad (5)$$

The structure shown in Fig. 1 is of a single patch RMSA (Rectangular Microstrip Antenna) which is inset fed by design.

Calculation for feedline:

The width of the feedline is an important parameter for designing single patch as well as for the array designing [5], [6].

$$A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{2} \left(0.23 + \frac{0.11}{\epsilon_r} \right) \quad (6)$$

$$B = \frac{377 * \pi}{2 * Z_0 \sqrt{\epsilon_r}} \quad (7)$$

Width of feedline:

$$W_f = \frac{2h}{\pi} \left\{ B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left(\ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right) \right\} \quad (8)$$

B. Quarter wave transformer:

The impedance for matching the 50Ω coaxial cable to the patch antenna and width and length of the transformer line is calculated in this session [3], [4]. Impedance of the patch can be given as,

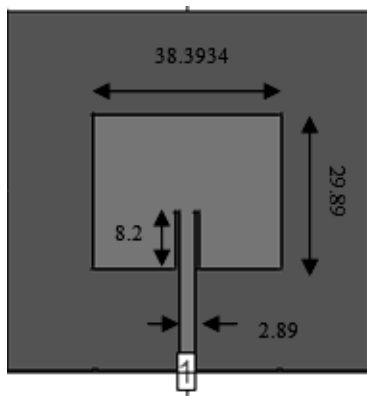


Figure 1. Patch antenna with inset feed

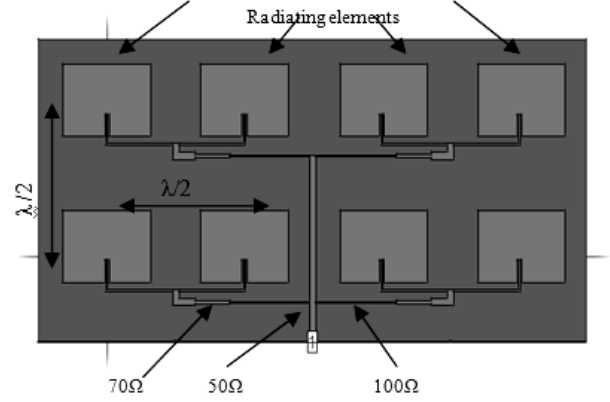


Figure 2. Patch array antenna with inset feed

$$Z_p = \frac{90 \epsilon_r^2}{\epsilon_r - 1} \left(\frac{L}{W} \right)^2 \quad (9)$$

$$y_o = \frac{L}{\pi} \cos^{-1} \sqrt{\frac{Z_0}{Z_1}} \quad (10)$$

where,

$$Z_1 = \sqrt{Z_o \times Z_p} \quad (11)$$

C. Microstrip 4x2 array design procedure:

As shown in Fig. 2, 4x2 array patch antenna designed using the configuration for single patch. All the above formulas are used to get the width and length dimension of the patch and also the feed lines. Quarter wave transformer of $\lambda/4$ length and impedance 70Ω is stuffed to match 100Ω and 50Ω lines. Distance between patches is kept as $\lambda/2$.

But for inset fed method the 50 Ω is divided into two 100 Ω microstrip lines thus dividing the power and phase equally forming uniform array. The 100 Ω and 50 Ω is matched by quarter wave transformer. The inset length y_o for the inset fed method is calculated using “(9), (10) and (11)”.

III. RESULTS AND DISCUSSIONS

Methods to increase gain can be acquired by changing the material or by use of superstrate, but by using simple design and changing patch parameters of array as well desired gain can be obtained. Many optimizations are performed by changing the width and length of the patches to obtain better values of radiating parameters as in Table II. The simulated results obtained better than the expected values are chosen to act as a good radiator which is highlighted in Table II. A single patch having $L=29.89\text{mm}$ and $W = 38.3934\text{mm}$ is observed to give $S_{11} = -10.14\text{dB}$ than calculated value of -3.74dB and $VSWR = 1.98$ rather than 2.46. Similarly, 4x2 array designed is optimized by length and width in which, $L=29.78\text{mm}$ and $W=38.3934\text{mm}$ is chosen to obtain $S_{11} = -$

19.64dB and VSWR=1.26. Comparisons can be made directly with respect to the radiating parameters which points

TABLE II: DESIGN SPECIFICATION:

Type of antenna		Patch width W (mm)	Patch Length L (mm)	S11 (dB)	VSWR	Directivity (dBi)	HPBW
Single Patch	From Calculation	38.3934	29.786	-3.744	2.46	-	-
	Length adjustment	38.3934	29.8	-8.01	2.4	6.74	82
		38.3934	29.9	-8.6	2.2	6.74	84
		38.3934	29.88	-10.08	1.99	6.74	82
		38.3934	29.89	-10.14	1.98	6.74	82
	Width adjustment	38.39	29.78	-6.11	3	6.74	82
		37	29.78	-9.73	2	6.74	82
		36.6	29.78	-9.99	2	6.74	82
		36.4	29.78	-10.05	1.99	6.74	82
	4x2	38.3934	29.786	-3.744	2.46	-	-
4x2	From Calculation	38.3934	29.786	-3.744	2.46	-	-
	Length adjustment	38.3934	29.78	-19.64	1.26	13.058	20
		38.3934	29.88	-11.08	1.77	12.53	24
		38.3934	29.9	-10.8	1.8	12.5	25
	Width adjustment	38	29.78	-9.5	2	12.08	30
		38.2	29.78	-9	2	12	32
		38.39	29.78	-9.2	2.07	13.14	30

out that the array antenna gives a directive and narrower beam than that of the single patch antenna. From Table II, it can be seen that HPBW of 20° is obtained which is narrower than 82° .

As shown in the Fig. 3 below, the S11 obtained for the array designed is -19.64dB which is best observed as per the optimized values in IE3D. The VSWR plot of the designed array is observed to be 1.26 as seen in the Fig. 4. And the bandwidth is around 0.24 GHz, which is far better than single patch antenna.

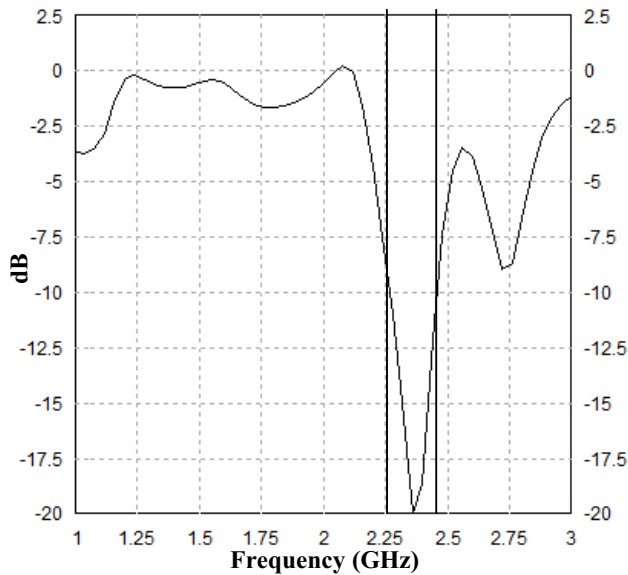


Figure 3. S11 of patch array antenna.

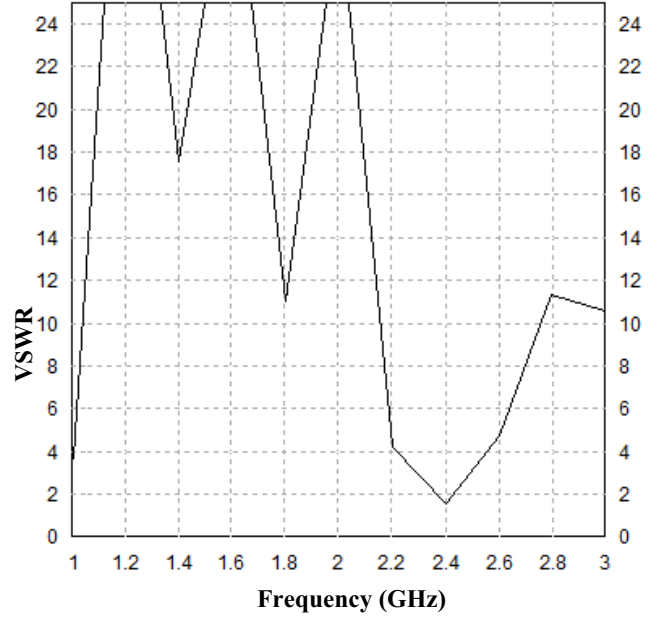


Figure 4. VSWR of patch array antenna

The Fig. 5 shown below describes the difference between gain of a single patch and 4x2 array. The diagram depicts that the array is more directive and has higher gain with respect to the single patch. The gain of array is observed to be 10.5dB and the directivity (D) is obtained as 13.058dBi as shown in Fig. 6.

The Half Power Beam Width (HPBW) obtained is also narrow for the array rather than the single patch RMSA which is 20° compared to 82° .

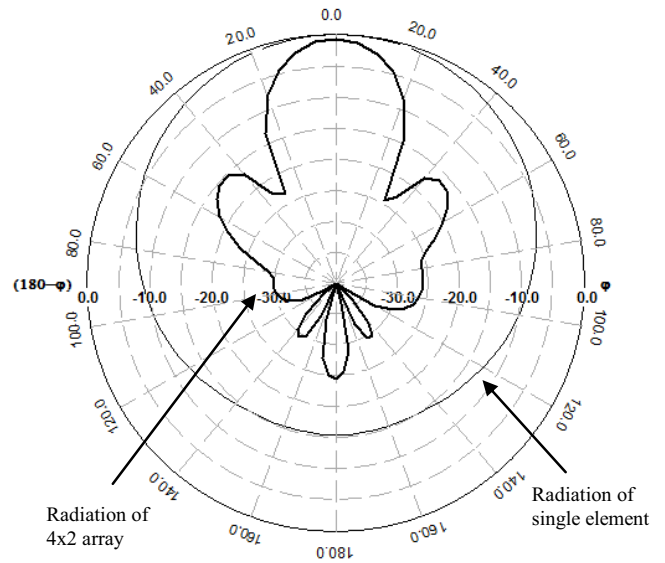


Figure 5. Comparison of Gain between single patch and 4x2 Array at 2.4GHz using IE3D

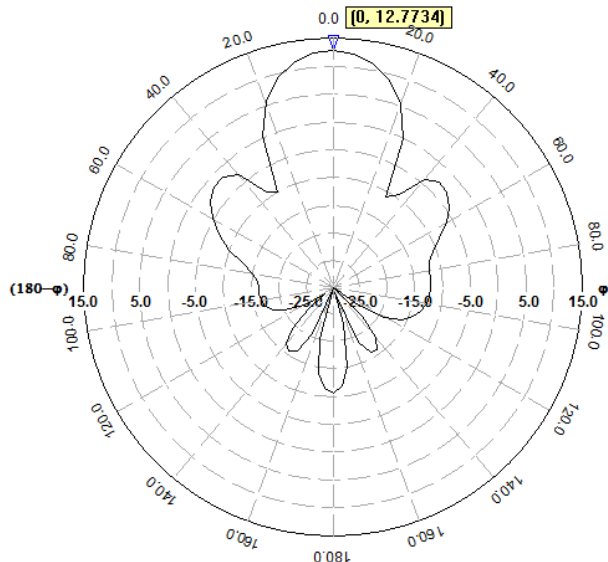


Figure 6. Directivity of patch array antenna at 2.4GHz using IE3D

Also a comparison can be made clearly from the next table, Table III that the selected optimization from [2] and simulations carried out here strikes that 4x2 yields better radiation parametric values such as gain, directivity etc., S11 and VSWR being close to that of 4x1. Also the HPBW is narrower in the case of 4x2 indicating more directivity.

The 3D diagram of radiation pattern of a single patch and the patch array of 4x2 designed as shown in the following Fig. 7 and Fig. 8.

TABLE III: COMPARISON TABLE [2]:

SR.NO	Type	L(mm)	W(mm)	Gain (dB)	S11 (dB)	D (dBi)	HPBW (deg)
1. Single Patch	2.4 GHz	38.393	29.89	3.7	-10.14	6.74	82
	2.5 GHz [2]	34.93	25.48	0.837	-15.4	5.8	NS
2. Array	4x2 (2.4GHz)	38.393	29.78	10.5	-19.64	13.05	20
	4x1 (2.5 GHz) [2]	34.93	25.1	5.732	-20.2	10.2	24.9

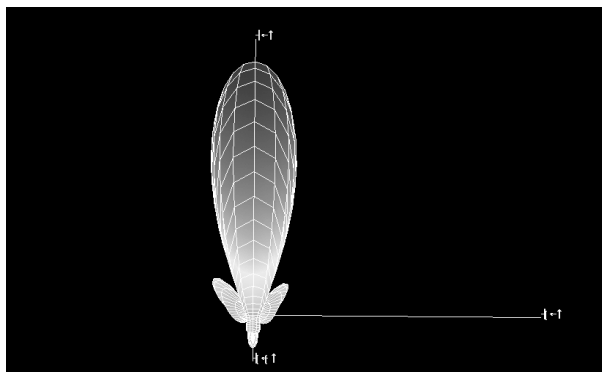


Figure 7. Radiation pattern of patch array antenna at 2.4 GHz

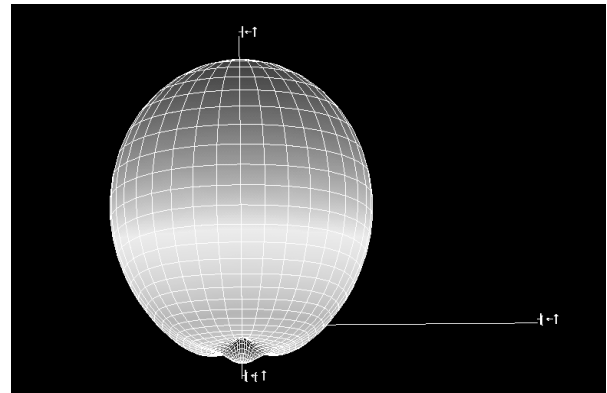


Figure 8. Radiation pattern of single patch at 2.4GHz

IV. CONCLUSION

A comparison is carried out between a single microstrip patch with the array of 4x2 (eight elements) in terms of important radiation parameters. The performance of the array outcomes the required results rather than the single patch for directivity, gain, return loss (S11) as well as the bandwidth. The desired requirements for 2.4 GHz is satisfied by 4x2 eight element patch array which gives return loss of about -19.64dB and a VSWR of 1.26 at the centre frequency of 2.4GHz. Additionally, the performance of the array is optimized by adjusting the length and width of each patch gives gain of 10.5dB relative to 3.7dB of single patch. It depends on other factors also such as feeding technique, substrate type, thickness of dielectric etc. respectively.

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